MODULE 5: COMPUTATIONAL CHEMISTRY: BASIC TO APPLICATIONS (CO: 5)

Complete Comprehensive Textbook: Quantum Mechanics, MO Theory, PES Minima & Saddle Points, DFT & AI Screening of Semiconductors

Aligned with Official VIT BACHY105 Applied Chemistry Course Syllabus, Department Question Bank & VIT FAT Papers

Syllabus Scope & Competency Checklist (Module 5):

- **Foundations of Quantum Chemical Calculations:** Postulates of quantum mechanics; Time-Independent Schrödinger Equation ($\hat{H}\psi = E\psi$); molecular Hamiltonian operator (electronic kinetic, nuclear kinetic, electron-nuclear attraction, electron-electron repulsion, nuclear-nuclear repulsion); Born-Oppenheimer approximation.
- **Molecular Orbital Theory (MOT):** Linear Combination of Atomic Orbitals (LCAO) framework; bonding (σ, π), anti-bonding (σ^*, π^*), and non-bonding orbitals; Aufbau, Pauli, and Hund rules; Bond Order formula (BO $= \frac{1}{2}(N_b - N_a)$).
- **MO Diagrams of Diatomic Molecules:**
  1. Homonuclear O_2 (16 electrons): Complete valence configuration $\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2p_z}^2\pi_{2p_x}^2\pi_{2p_y}^2\pi_{2p_x}^{*1}\pi_{2p_y}^{*1}$; bond order $= 2.0$; paramagnetic behavior with 2 unpaired electrons in degenerate π^* orbitals.
  2. Heteronuclear CO (14 electrons): Electronegativity asymmetry (oxygen orbitals lowered relative to carbon); $1\sigma^2 2\sigma^2 1\pi^4 3\sigma^2$; bond order $= 3.0$; HOMO (3σ carbon lone pair) and LUMO (degenerate $2\pi^*$); π-backbonding in transition metal carbonyl complexes.
- **Potential Energy Surfaces (PES):** $3N-6$ dimensional nuclear coordinate space; Gradient vector (∇V); Hessian force constant matrix ($\partial^2 \frac{V}{\partial} q_i \partial q_j$); Local and Global Minima (all positive eigenvalues, zero imaginary frequencies); Transition States (first-order saddle points, exactly one negative eigenvalue, one imaginary frequency along Intrinsic Reaction Coordinate IRC).
- **Intermolecular Potentials:** Lennard-Jones (12-6) potential ($V_{\mathrm{LJ}(r)} = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$); Pauli repulsive core vs long-range London dispersion attraction; potential well depth (ε) and equilibrium separation ($r_m = 2^{\frac{1}{6}}\sigma$).
- **Geometry Optimization & Harmonic Frequencies:** Energy minimization algorithms (Steepest Descent, Conjugate Gradient, BFGS); harmonic vibrational frequencies, Zero-Point Vibrational Energy (ZPVE); thermochemical corrections ($\Delta H, \Delta G$).
- **Density Functional Theory (DFT) & AI Screening:** Hohenberg-Kohn theorems ($E = E[\rho]$); Kohn-Sham equations; exchange-correlation functionals (LDA, GGA PBE, Hybrid B3LYP); Artificial Intelligence and Graph Neural Networks (CGCNN) for rapid high-throughput screening of semiconductor bandgaps and battery solid electrolytes.

1. Quantum Mechanical Calculations & Born-Oppenheimer Approximation

1.1 The Molecular Schrödinger Equation

All computational chemistry methods originate from the fundamental Time-Independent Schrödinger Equation:

$$\hat{H}\psi(\boldsymbol{r}, \boldsymbol{R}) = E\psi(\boldsymbol{r}, \boldsymbol{R})$$

where $\boldsymbol{r}$ denotes electronic coordinates, $\boldsymbol{R}$ denotes nuclear coordinates, ψ is the molecular wavefunction, and E is the total energy. The complete non-relativistic molecular Hamiltonian operator $\hat{H}$ consists of five distinct energetic contributions:

$$\hat{H} = -\sum_A \frac{\text{planck.reduce}^2}{2M_A}\nabla_A^2 - \sum_i \frac{\text{planck.reduce}^2}{2m_e}\nabla_i^2 - \sum_A \sum_i \frac{Z_A e^2}{4\pi\varepsilon_0 r_{Ai}} + \sum_{i<j} \frac{e^2}{4\pi\varepsilon_0 r_{ij}} + \sum_{A<B} \frac{Z_A Z_B e^2}{4\pi\varepsilon_0 R_{AB}}$$

- **Terms:**
  1. $-\sum_A \frac{\text{planck.reduce}^2}{2M_A}\nabla_A^2$: Kinetic energy of the atomic nuclei.
  2. $-\sum_i \frac{\text{planck.reduce}^2}{2m_e}\nabla_i^2$: Kinetic energy of the electrons.
  3. $-\sum_A \sum_i \frac{Z_A e^2}{4\pi\varepsilon_0 r_{Ai}}$: Attractive Coulomb potential between positively charged nuclei and electrons.
  4. $+\sum_{i<j} \frac{e^2}{4\pi\varepsilon_0 r_{ij}}$: Repulsive Coulomb potential between electrons (electron-electron repulsion).
  5. $+\sum_{A<B} \frac{Z_A Z_B e^2}{4\pi\varepsilon_0 R_{AB}}$: Repulsive Coulomb potential between atomic nuclei.

1.2 The Born-Oppenheimer Approximation (1927)

Because an atomic nucleus (proton) is at least 1836 times heavier than an electron ($M_A \gg m_e$), electrons move orders of magnitude faster than nuclei.

- **Physical Formulation:** We can clamp the nuclei at fixed spatial positions ($\boldsymbol{R} = \text{constant}$). The nuclear kinetic energy term vanishes, and nuclear repulsion becomes a constant electrostatic shift.
- **Electronic Schrödinger Equation:**

$$\hat{H}_{\text{elec}}\psi_{\text{elec}(\boldsymbol{r};\boldsymbol{R})} = E_{\text{elec}(\boldsymbol{R})}\psi_{\text{elec}(\boldsymbol{r};\boldsymbol{R})}$$

Total molecular energy at fixed nuclear geometry:

$$E_{\text{total}(\boldsymbol{R})} = E_{\text{elec}(\boldsymbol{R})} + V_{\text{NN}(\boldsymbol{R})}$$

Solving $E_{\text{total}(\boldsymbol{R})}$ for all possible nuclear arrangements maps out the **Potential Energy Surface (PES)**.

2. Molecular Orbital Theory: MO Diagrams of O_2 & CO

2.1 The LCAO Approximation & Bond Order

Molecular orbitals are constructed mathematically as a Linear Combination of Atomic Orbitals (LCAO):

$$\psi_{\text{MO}} = c_1\varphi_1 \text{ pm } c_2\varphi_2$$

- **Bonding Orbital ($\boldsymbol{\psi_b = c_1\varphi_1 + c_2\varphi_2}$):** In-phase constructive interference increases electron probability density between nuclei, lowering electronic energy (σ, π).
- **Antibonding Orbital ($\boldsymbol{\psi_a = c_1\varphi_1 - c_2\varphi_2}$):** Out-of-phase destructive interference creates a nodal plane between nuclei, raising electronic energy (σ^*, π^*).
- **Bond Order Formulation:**

$$\text{Bond Order (BO)} = \frac{1}{2}(N_b - N_a)$$

where N_b is the number of electrons in bonding MOs and N_a is the number of electrons in antibonding MOs. Higher bond order corresponds to shorter bond length and higher bond dissociation energy.

2.2 MO Energy Level Diagram of Homonuclear O_2 (Oxygen Molecule)

Oxygen has atomic number $Z = 8$; an O_2 molecule possesses 16 total electrons (12 valence electrons):

- **Valence Molecular Orbital Electronic Configuration:**

$$\sigma_{2s}^2 \quad \sigma_{2s}^{*2} \quad \sigma_{2p_z}^2 \quad \left(\pi_{2p_x}^2 = \pi_{2p_y}^2\right) \quad \left(\pi_{2p_x}^{*1} = \pi_{2p_y}^{*1}\right)$$

- **Bond Order Calculation:**
  - Bonding electrons (N_b): 2 in σ_{2s}, 2 in σ_{2p_z}, 4 in $\pi_{2p} = 8$ electrons.
  - Antibonding electrons (N_a): 2 in σ_{2s}^*, 2 in $\pi_{2p}^* = 4$ electrons.

$$\text{Bond Order} = \frac{1}{2}(8-4) = \frac{4}{2} = 2.0 \quad (\text{Stable Double Bond} \;\; O=O)$$

- **Paramagnetism of $\boldsymbol{O_2}$:** By Hund's rule of maximum multiplicity, the two highest-energy electrons enter the two degenerate antibonding orbitals ($\pi_{2p_x}^*$ and $\pi_{2p_y}^*$) with **parallel unpaired spins** ($\uparrow, \uparrow$). This rigorously explains why liquid oxygen is strongly attracted into magnetic fields (**paramagnetic**), resolving the historical failure of Lewis and Valence Bond theories which incorrectly predicted complete spin pairing (diamagnetism).

2.3 MO Diagram of Heteronuclear CO (Carbon Monoxide)

Carbon Monoxide possesses 14 total electrons (10 valence electrons, isoelectronic with N_2). Because oxygen is significantly more electronegative than carbon ($\chi_O = 3.44 > \chi_C = 2.55$), oxygen's atomic $2s$ and $2p$ orbitals lie lower in energy than carbon's:

- **Molecular Orbital Configuration:**

$$1\sigma^2 \quad 2\sigma^2 \quad 1\pi^4 \quad 3\sigma^2$$

- **Bond Order Calculation:**

$$\text{Bond Order} = \frac{1}{2}(8-2) = \frac{6}{2} = 3.0 \quad (\text{Extremely Strong Triple Bond} \;\; C \text{ equiv } O)$$

- **Frontier Molecular Orbitals & Metal Carbonyl Bonding:**

- **HOMO (3σ orbital):** Formed primarily from carbon's $2p_z$ orbital, directed outward as a localized non-bonding lone pair on the Carbon atom (not oxygen!). Carbon acts as a Lewis σ-donor to transition metal d-orbitals.
- **LUMO (Degenerate $2\pi^*$ orbitals):** Empty, low-lying antibonding orbitals localized predominantly on the Carbon atom. Transition metal d-electrons back-donate into CO's $2\pi^*$ LUMO (**π-backbonding**), strengthening the metal-carbon bond while weakening the $C-O$ bond (shifting the $C-O$ IR stretching frequency downward from $2143\ \text{cm}^{-1}$).

3. Potential Energy Surfaces (PES) & Intermolecular Potentials

3.1 Topography of the Potential Energy Surface (PES)

For a non-linear molecule of N atoms, the PES is a $(3N-6)$ dimensional hypersurface representing potential energy V as a function of nuclear geometric coordinates $\boldsymbol{q} = (q_1, q_2, ..., q_{3N-6})$:

- **Gradient Vector (First Derivatives):**

$$\boldsymbol{g} = \nabla V = \begin{pmatrix} \partial \frac{V}{\partial} q_1 \\ \partial \frac{V}{\partial} q_2 \\ \text{vdots} \\ \partial \frac{V}{\partial} q_{3N-6} \end{pmatrix}$$

At all **Stationary Points**, the gradient vanishes identically: $\nabla V = \mathbf{0}$.

- **Hessian Matrix (Second Derivatives & Force Constants):**

$$\boldsymbol{H}_{ij} = \frac{\partial^2 V}{\partial q_i \partial q_j}$$

Diagonalizing the mass-weighted Hessian matrix yields the fundamental harmonic vibrational frequencies (ν_i prop $\sqrt{\lambda_i}$):

1. **Local / Global Minima:** All eigenvalues (λ_i) of the Hessian matrix are **strictly positive** ($\lambda_i > 0$). All vibrational frequencies are real numbers ($\nu_i > 0$, **zero imaginary frequencies**). Represents stable physical chemical species (reactants, intermediates, products).
2. **Transition States (First-Order Saddle Points):** Exactly **one eigenvalue is strictly negative** ($\lambda_1 < 0$) while all remaining $(3N-7)$ eigenvalues are positive ($\lambda_i > 0$).
   - The negative eigenvalue corresponds to **exactly one imaginary vibrational frequency** ($\nu_1 = i\ |\nu_1|$).
   - The corresponding normal mode eigenvector defines the **Intrinsic Reaction Coordinate (IRC)**, representing the minimum energy path crossing the activation barrier connecting reactants to products.

3.2 Intermolecular Potentials: The Lennard-Jones (12-6) Model

The Lennard-Jones potential describes the interaction energy between two non-bonding neutral atoms or molecules separated by distance r:

$$V_{\text{LJ}(r)} = 4\varepsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right]$$

- **Physical Origin of Terms:**
  1. **Repulsive Term ($\left(\frac{\sigma}{r}\right)^{12}$):** Dominates at very short separations ($r < \sigma$). Arises from Pauli repulsion when occupied electron clouds overlap.
  2. **Attractive Term ($-\left(\frac{\sigma}{r}\right)^{6}$):** Dominates at intermediate and long distances. Arises from quantum mechanical induced dipole-induced dipole London dispersion interactions.
- **Equilibrium Characteristics:**
  - Zero-potential crossing point: $V_{\text{LJ}(\sigma)} = 0$.
  - Minimum energy (well depth): Occurs by setting $d\frac{V_{\text{LJ}}}{d}r = 0$:

$$\frac{d}{dr} V_{\text{LJ}(r)} = 4\varepsilon \left[-12\frac{\sigma^{12}}{r^{13}} + 6\frac{\sigma^6}{r^7} \right] = 0 \quad \Longrightarrow \quad 12\frac{\sigma^{12}}{r^{13}} = 6\frac{\sigma^6}{r^7}$$

$$r^6 = 2\sigma^6 \quad \Longrightarrow \quad r_m = 2^{\frac{1}{6}}\sigma \approx 1.12246\sigma$$

Substituting r_m back into potential:

$$V_{\mathrm{LJ}(r_m)} = 4\varepsilon\left[\left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)\right] = 4\varepsilon\left[\frac{1}{4} - \frac{1}{2}\right] = -\varepsilon$$

The potential well depth is identically $-\varepsilon$, representing the bond dissociation energy of the van der Waals dimer.

4. Density Functional Theory (DFT) & AI Screening of Semiconductors

4.1 Foundations of Density Functional Theory (DFT)

Traditional wave-function methods (Hartree-Fock, Coupled Cluster) scale exponentially with electron count because an N-electron wavefunction depends on $3N$ spatial coordinates.

- **Hohenberg-Kohn Theorems (1964):**
  1. **Theorem 1:** The ground-state electronic energy and all physical properties of a many-electron system are uniquely and completely determined by the three-dimensional ground-state electron density $\rho(\boldsymbol{r})$:

$$E = E[\rho(\boldsymbol{r})]$$

  reducing $3N$ spatial variables to just 3 spatial dimensions (x, y, z).
  2. **Theorem 2:** The true ground-state electron density minimizes the total energy functional (variational principle).
- **Kohn-Sham Equations (1965):** Map the intractable interacting electron system onto an equivalent fictitious non-interacting system yielding the same electron density:

$$\left[-\frac{\mathrm{planck.reduce}^2}{2m_e}\nabla^2 + V_{\mathrm{eff}(\boldsymbol{r})}\right]\psi_{i(\boldsymbol{r})} = \varepsilon_i\psi_{i(\boldsymbol{r})}$$

  where effective potential $V_{\mathrm{eff}} = V_{\mathrm{ext}} + V_{\mathrm{Hartree}} + V_{\mathrm{xc}}$. The unknown physics is isolated in the **Exchange-Correlation Functional ($E_{\mathrm{xc}[\rho]}$)**:
  - **LDA (Local Density Approximation):** E_{xc} depends only on local density $\rho(\boldsymbol{r})$.
  - **GGA (Generalized Gradient Approximation, e.g., PBE):** Depends on density and its gradient $(\nabla\rho)$.
  - **Hybrid Functionals (e.g., B3LYP, HSE06):** Incorporates a fraction of exact Hartree-Fock exchange, providing accurate semiconductor bandgap predictions.

4.2 AI & Machine Learning for Semiconductor Screening

Screening hundreds of thousands of candidate materials using pure DFT is computationally prohibitive ($10 - 100$ CPU hours per crystal).

- **Machine Learning Pipeline:**
  1. **Crystal Graph Representation (CGCNN):** Atoms are represented as graph nodes (encoded with atomic number, electronegativity, radius); interatomic bonds are represented as edges (encoded with bond distances).
  2. **Deep Graph Neural Networks:** Predict formation energy (ΔH_f), bandgap (E_g), carrier mobilities (μ), and structural stability in milliseconds.
  3. **Applications:** Discovering stable lead-free halide perovskites ($\mathrm{Cs_2AgBiBr_6}$) for solar cells and superionic lithium conductors ($\mathrm{Li_{10}GeP_2S_{12}}$) for all-solid-state batteries.

VIT FAT Solved Question Bank: Step-by-Step Derivations

FAT Problem 1: Molecular Orbital Diagram & Magnetic Properties of O_2 vs CO
Question (10 Marks): ("i") Construct the complete valence molecular orbital energy level diagram for the homonuclear diatomic molecule O_2 and heteronuclear CO. ("ii") Write their full electronic configurations and calculate the bond order for both molecules. ("iii") Explain why liquid O_2 is paramagnetic while CO is diamagnetic based on molecular orbital theory. ("iv") When O_2 is ionized to O_2^+ and O_2^-, predict how their respective bond lengths and bond dissociation energies change relative to neutral O_2.

Complete Step-by-Step Solution:
Step 1: Configurations and Bond Orders:
- **O_2 (12 Valence Electrons):**

$$\sigma_{2s}^2 \quad \sigma_{2s}^{*2} \quad \sigma_{2p_z}^2 \quad \pi_{2p_x}^2 \quad \pi_{2p_y}^2 \quad \pi_{2p_x}^{*1} \quad \pi_{2p_y}^{*1}$$

$$\text{BO}(O_2) = \frac{1}{2}(N_b - N_a) = \frac{1}{2}(8-4) = 2.0 \quad \text{(Double Bond)}$$

- **CO (10 Valence Electrons):**

$$1\sigma^2 \quad 2\sigma^2 \quad 1\pi^4 \quad 3\sigma^2$$

$$\text{BO(CO)} = \frac{1}{2}(8-2) = 3.0 \quad \text{(Triple Bond)}$$

Step 2: Magnetic Property Explanation:
- In O_2, the two highest energy electrons occupy two degenerate antibonding orbitals ($\pi_{2p_x}^*, \pi_{2p_y}^*$) with parallel spins ($\uparrow, \uparrow$) according to Hund's rule. The presence of two unpaired electrons gives O_2 a net permanent magnetic dipole moment ($\mu = \sqrt{n(n+2)} = \sqrt{2(4)} = 2.83\mu_B$), making it strongly **paramagnetic**.
- In CO, all 14 electrons are completely paired in closed shells with zero unpaired electrons ($n = 0, \mu = 0$), making it strictly **diamagnetic**.

Step 3: Ionization Effects on O_2^+ and O_2^-:
- **Cation O_2^+:** Removing one electron from an antibonding π^* orbital decreases N_a from 4 to 3:

$$\text{BO}(O_2^+) = \frac{1}{2}(8-3) = 2.5$$

  Higher bond order ($2.5 > 2.0$) results in a **shorter bond length** and **higher bond dissociation energy**!
- **Anion O_2^- (Superoxide):** Adding one electron into an antibonding π^* orbital increases N_a from 4 to 5:

$$\text{BO}(O_2^-) = \frac{1}{2}(8-5) = 1.5$$

  Lower bond order ($1.5 < 2.0$) results in a **longer bond length** and **lower bond dissociation energy**!

Boxed Final Answers:

$$\text{BO}(O_2) = 2.0, \quad \text{BO(CO)} = 3.0, \quad O_2 \text{ has 2 unpaired } \pi^* \text{ electrons (paramagnetic)}, \quad \text{BO}(O_2^+) = 2.5, \text{BO}(O_2^-) = 1.5.$$

FAT Problem 2: Lennard-Jones Potential Parameters & Equilibrium Force Constants
Question (10 Marks): For Argon gas, the empirical Lennard-Jones (12-6) potential parameters are determined from transport measurements to be:

$$\varepsilon = 1.656 \times 10^{-21} \text{ Joules} \quad \left(\frac{\varepsilon}{k_B} = 120 \text{ K}\right), \quad \sigma = 3.40 \text{ \u00c5} = 0.340 \text{ nm}$$

("i") Calculate the equilibrium interatomic separation distance r_m where the potential energy is minimum. ("ii") Compute the value of the potential energy at $r = \sigma$, $r = r_m$, and $r = 2.5\sigma$. ("iii") Derive the mathematical expression for the force $F(r) = -d\frac{V}{d}r$ between two Argon atoms and verify that $F(r_m) = 0$. ("iv") Calculate the effective harmonic force constant $k = \left(d^2\frac{V}{d}r^2\right)|_{r=r_m}$ at the bottom of the potential well.

Complete Step-by-Step Solution:
Step 1: Calculate Equilibrium Separation r_m:

$$r_m = 2^{\frac{1}{6}}\sigma = (1.12246) \times (3.40 \text{ \u00c5}) = 3.816 \text{ \u00c5} = 0.3816 \text{ nm}$$

Step 2: Potential Energy Values:
- At $r = \sigma$: $V_{\text{LJ}(\sigma)} = 4\varepsilon[1^{12} - 1^6] = 0$ Joules.
- At $r = r_m$: $V_{\text{LJ}(r_m)} = -\varepsilon = -1.656 \times 10^{-21}$ Joules $= -0.01034$ eV.
- At $r = 2.5\sigma$:

$$\left(\frac{\sigma}{r}\right)^6 = \left(\frac{1}{2.5}\right)^6 = (0.40)^6 = 4.096 \times 10^{-3}$$

$$\left(\frac{\sigma}{r}\right)^{12} = \left(4.096 \times 10^{-3}\right)^2 = 1.678 \times 10^{-5}$$

$$V_{\text{LJ}(2.5\sigma)} = 4\varepsilon[1.678 \times 10^{-5} - 4.096 \times 10^{-3}] \approx 4 \times \left(1.656 \times 10^{-21}\right) \times \left(-4.079 \times 10^{-3}\right) = -2.70 \times 10^{-23} \text{ J}$$

.

Step 3: Force Derivation:

$$F(r) = -\frac{dV_{\text{LJ}}}{dr} = -4\varepsilon\left[-12\frac{\sigma^{12}}{r^{13}} + 6\frac{\sigma^6}{r^7}\right] = \frac{24\varepsilon}{r}\left[2\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^6\right]$$

At $r = r_m = 2^{\frac{1}{6}}\sigma$:

$$\left(\frac{\sigma}{r_m}\right)^6 = \frac{1}{2} \implies 2\left(\frac{\sigma}{r_m}\right)^{12} = 2\left(\frac{1}{4}\right) = \frac{1}{2}$$

$$F(r_m) = \frac{24\varepsilon}{r_m}\left[\frac{1}{2} - \frac{1}{2}\right] = 0 \text{ N (Equilibrium).}$$

Step 4: Harmonic Force Constant k:

$$k = \frac{d^2V}{dr^2}|_{r_m} = \frac{72\varepsilon}{2^{\frac{1}{3}}\sigma^2} = \frac{72 \times \left(1.656 \times 10^{-21}\right)}{1.2599 \times \left(3.40 \times 10^{-10}\right)^2} = \frac{1.1923 \times 10^{-19}}{1.2599 \times 1.156 \times 10^{-19}} = 0.8186 \text{ N/m}$$

Boxed Final Answers:

$$r_m = 3.82 \text{ \u00c5}, \quad V(\sigma) = 0, \quad V(r_m) = -1.656 \times 10^{-21} \text{ J}, \quad F(r_m) = 0, \quad k = 0.819 \text{ N/m}.$$